AI 321: Machine Learning for Sequential Data

System Models of Time-Series

Rhandley D. Cajote, PhD

Electrical and Electronics Engineering Institute (EEEI)
University of the Philippines Diliman

August 28, 2024

Overview

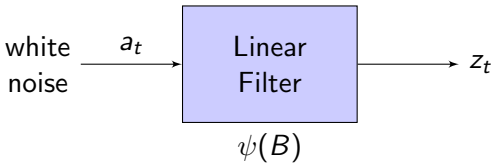
1. Linear Filter Models
2. Autoregressive Models (AR)
3. Moving Average Models (MA)
4. Autoregressive Moving Average Models (ARMA)
5. Nonstationary models
6. Transfer Function Model
7. Models with Superimposed Noise

Linear Filter Model [1]

- Observable time series z_t with highly dependent values can be generated from a series of *independent* "shocks" a_t.
- The "shocks" are random samples drawn from a distribution, with zero mean and variance σ_a.
- The sequence of independent random variables is called *white noise* process

$$a_t, a_{t-1}, a_{t-2}, \ldots$$

- the linear filter transforms that white noise process a_t to the process z_t



Linear Filter Model

- The linear filtering operation takes the weighted sum of previous shocks a_t

$$\begin{aligned} z_t &= \mu + a_t + \psi_1 a_{t-1} + \psi_2 a_{t-2} + \cdots \\ &= \mu + \psi(B) a_t \end{aligned}$$

- In general. μ is a parameter that determines the "level" of the process, and
- The operator, $\psi(B)$, is defined as:

$$\psi(B) = 1 + \psi_1 B + \psi_2 B^2 + \cdots$$

- The linear operator $\psi(B)$ is called the *transfer function* of the filter.

Impulse Response of the Filter

- The sequence formed by the weights can be finite or infinite

$$\psi_1, \psi_2, \psi_3, \ldots$$

- They are the filter *coefficients* or the impulse response of the filter (FIR or IIR)
- If the sequence is *absolutely summable*.

$$\sum_{j=0}^{\infty} |\psi_j| < \infty$$

  the filter is said to be *stable* and the process z_t is stationary
- The parameter μ is the mean of the process
- Otherwise, z_t is non-stationary and μ has no meaning.

Regression Model

- A linear model defined as

$$\tilde{z} = \phi_1 \tilde{x_1} + \phi_2 \tilde{x_2} + \cdots + \phi_p \tilde{x_p} + a$$

- relates a "dependent" variable z to a set of "independent" variables $x_1, x_2, \ldots, x_p$ plus a random error a is referred to as a *regression model*.
- z is said to be "regressed" on $x_1, x_2, \ldots, x_p$

Autoregressive Models

- The current value of the process is a linaar combination of the *previous values* of the process and a random shock a_t
- The values of a process at equally spaced time $t, t-1, t-2, \ldots$ be $z_t, z_{t-1}, z_{t-1}, \ldots$.
- Let $\tilde{z}_t = z_t - \mu$, be a series of deviations from μ, then

$$\tilde{z}_t = \phi_1 \tilde{z}_{t-1} + \phi_2 \tilde{z}_{t-2} + \cdots + \phi_p \tilde{z}_{t-p} + a_t$$

- is called the *autoregressive (AR) process of order p*
- The variable z is regressed on previous values of itself, hence *auto*regressive.
- The autoregressive operator of order p is defined:

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p$$

- written concisely as

$$\phi(B)\tilde{z}_t = a_t$$

Autoregressive Models

- The autoregressive model written as

$$\phi(B)\tilde{z}_t = a_t$$

  contains $p + 2$ parameters $\mu, \phi_1, \phi_2, \ldots, \phi_p, \sigma_a^2$. σ_a^2 is the variance of the white noise process a_t. In practice these parameters are estimated from the data
- *finite* weighted sum of p previous deviations $\tilde{z}_{t-1}, \tilde{z}_{t-2}, \ldots, \tilde{z}_{t-p}$ of the process plus a random shock a_t
- is equivalent to (special case of a linear model)

$$\tilde{z}_t = \phi^{-1}(B)a_t = \psi(B)a_t$$

  with $\psi(B) = \phi^{-1}(B) = \sum_{j=0}^{\infty} \psi_j B^j$
- *infinite* weighted sum of a's

Moving Average Models

- finite *moving average* process, $\tilde{z}_t$ linearly dependent on a finite number q of previous a's.

$$\tilde{z}_t = a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2} - \cdots - \theta_q a_{t-q}$$

  is called a *moving average* (MA) *process of order q*.

- We define a *moving average operator of order q* as:

$$\theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q$$

- written economically as

$$\tilde{z}_t = \theta(B) a_t$$

  contains $q + 2$ parameters $\mu, \theta_1, \ldots, \theta_q, \sigma_a^2$, which in practice is estimated from the data

Autoregressive-Moving Average Models

- To achieve greater flexibility in fitting of actual time series, combine AR and MA models.

$$\tilde{z}_t = \phi_1 \tilde{z}_{t-1} + \cdots + \phi_p \tilde{z}_{t-p} + a_t - \theta_1 a_{t-1} - \cdots - \theta_q a_{t-q}$$

or

$$\phi(B)\tilde{z}_t = \theta(B)a_t$$

- The model has $p + q + 2$ unknown parameters $\mu, phi_1, \ldots, \phi_p, \theta_1, \ldots, \theta_q, \sigma_a * 2$ that are estimated from data.
- Can be written in the form of a linear filter

$$\tilde{z}_t = \phi^{-1}(B)\theta(B)a_t = \psi(B)a_t$$

with $\psi(B) = \phi^{-1}(B)\theta(B)$.

Non-stationary models

- Non-stationary with homogeneous behavior over time can be models with AR, MA and ARMA models
- Autoregressive integrated Moving Average (ARIMA) process can model homogeneous non-stationary behavior
- Created by "integrating" or "summing" the stationary ARMA process
- Reading assignment

Transfer Function Model

- For discrete data, the transfer relationship between output Y and input X define by a difference equation

$$(1 + \xi_1 \nabla + \cdots + \xi_r \nabla^r) Y_t = (\eta_0 + \eta_1 \nabla + \cdots + \eta_s \nabla^t) X_{t-b}$$

  in which the differential operator $D = d/dt$ is replaced with a difference operator $\nabla = 1 - B$

- The model with few parameters ($r \leq 2, s \leq 2$), may often be used to approximate the dynamic relationship whose true nature is more complex.

Transfer Function Model

- In terms of past values of input and output, $B = 1 - \nabla$

$$\begin{aligned}(1 - \delta_1 B - \cdots - \delta_t B^r) Y_t &= (\omega_0 - \omega_1 B - \cdots - \omega_s B^s) X_{t-b} \\ &= (\omega_0 B^b - \omega_1 B^{b+1} - \cdots - \omega_s B^{b+s}) X_t\end{aligned}$$

  or

$$\delta(B) Y_t = \omega(B) B^b X_t = \Omega(B) X_t$$

- Output Y_t and input X_t are linked by a linear filter

$$\begin{aligned}Y_t &= \nu_0 X_t + \nu_1 X_{t-1} + \nu_2 X_{t-2} + \cdots \\ &= \nu(B) X_t\end{aligned}$$

- The transfer function (impulse response coefficients)

$$\nu(B) = \nu_0 + \nu_1 B + \nu_2 B^2 + \cdots$$

Transfer Function Model

- The transfer function can be expressed as a ratio of two polynomials

$$\nu(B) = \frac{\Omega(B)}{\delta(B)} = \delta^{-1}(B)\Omega(B)$$

- The sequence of weights $\nu_0, \nu_1, \nu_2 \ldots$ is called the *impulse response function*
- The first b weights ν_0, ν_1, ν_{b-1} are zero
- The filter is *stable* if the coefficients $\{\nu_j\}$ are absolutely summable

$$\sum_{j=0}^{\infty} |\nu_j| < \infty$$

Models with Noise

- In the presence of noise, the model becomes:

$$Y_t = \nu(B)X_t + N_t$$

where N_t and X_t are independent processes, N_t can be stationary or non-stationary.

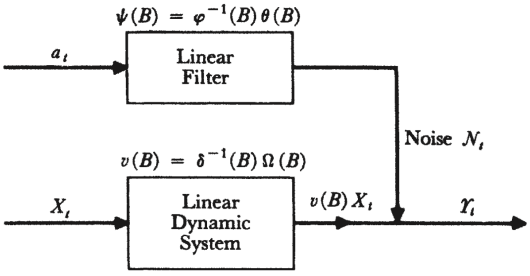


Figure: Transfer function of LDS + Noise

Models with Noise

- We have
$$N_t = \psi(B)a_t = \varphi^{-1}(B)\theta(B)a_t$$
- The input and output will be
$$\begin{aligned} Y_t &= \nu(B)X_t + \psi(B)a_t \\ &= \delta^{-1}(B)\Omega(B)X_t + \varphi^{-1}(B)\theta(b)a_t \end{aligned}$$
- In practice it is necessary to estimate the transfer function of the linear filter describing the noise $\psi(B)$ and also the transfer function $\nu(B)$

References

[1] G. E. P. Box and G. Jenkins, *Time Series Analysis, Forecasting and Control*, 4th ed. USA: John Wiley and Sons, 2008, 2008, ISBN: 9780470272848.